

ABDULLAH AL MARUF

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RESEARCH INTERESTS

Vision-language models; multimodal reasoning and visual grounding; reliable and efficient multimodal learning; knowledge distillation; robustness, evaluation, and evidence-grounded reasoning.

EDUCATION

Rajshahi University of Engineering & Technology (RUET)

B.Sc. in Computer Science & Engineering | CGPA: 3.55/4.00 after six semesters

Rajshahi, Bangladesh

Expected Mar 2027

PUBLICATIONS AND MANUSCRIPTS

Peer-reviewed workshop paper

- **Abdullah Al Maruf**, Md. Sajedul Islam, Tanmoy Mridha, and Irfan Hossain Bhuiyan. “BD-HazardVLM: Probing Vision-Language Models for Latent Defensive-Driving Hazards in Bangladesh Road Scenes.” *ECCV 2026 Workshop on Safe and Defensive Autonomous Driving (SDAD)*, accepted as a poster.

Manuscripts under review

- **Abdullah Al Maruf**. “DINOv2 Feature Distillation for Recurrent Local-Update Image Classifiers.” Submitted to the LIGHT Workshop at NeurIPS 2026.
- Ushashi Bhattacharjee, **Abdullah Al Maruf**, Koushik Howlader, Sayantan Chakraborty, and Tirtho Roy. “Visual Reference Tables Recover Signal That Tissue Captions Alone Miss.” Submitted to the MedReasoner Workshop at NeurIPS 2026.
- **Abdullah Al Maruf** and Tirtho Roy. “Single-Pass Uncertainty for Selective Polyp Segmentation under External Dataset Variation.” Submitted to the ML4H 2026 Symposium.

RESEARCH EXPERIENCE

Undergraduate Thesis – BanglaNewsGround

Supervisor: Md. Nasif Osman Khansur, Assistant Professor, Department of CSE, RUET

Sep 2026 – Present

- Study how caption supervision shapes Bengali vision-language representations by comparing original journalist captions, text-only rewrites, visually grounded recaptions, and combined supervision.
- Design controlled experiments on approximately 30,000 Bengali news images, including a human-grounded internal test set and cross-architecture evaluation across multiple encoder families.
- Evaluate image-text retrieval, image-caption matching, and hard semantic/entity negatives to separate genuine visual-grounding gains from improvements caused by cleaner language.

Independent Research – Tiny VLM Relational Reasoning

2026 – Present

- Investigate whether sub-billion-parameter vision-language models acquire reusable visual relations or rely on linguistic and multimodal shortcuts.
- Design verified, question-conditioned structured distillation with answer-only, unstructured-rationale, relational-supervision, and corrupted-relation controls.
- Use counterfactual pair evaluation and visual-dependence controls to test causal use of relations while targeting direct-answer inference at deployment.

Independent Research – Spec2Inspect-Lite: Evidence-Grounded Industrial Inspection

2026 – Present

- Develop an auditable MVTec AD inspection pipeline that converts localized anomaly evidence into explicit measurements and specification-grounded decisions rather than unconstrained VLM judgments.
- Compare direct Qwen judgments with structured component/evidence fusion, oracle-evidence upper bounds, and specification-only shortcut controls across cable, capsule, and metal_nut categories.
- Design threshold, relation, and zone counterfactuals to test whether decisions respond to task-relevant visual evidence while remaining stable to irrelevant changes.

SELECTED RESEARCH CONTRIBUTIONS

BD-HazardVLM

2026

- Developed a 500-image Bangladesh road-scene benchmark for latent defensive-driving hazard reasoning, with annotations for visible evidence, hazard category, risk timing, severity, defensive actions, and hallucination/bias traps.
- Evaluated Qwen2.5-VL-7B, InternVL3-8B, and MiniCPM-V-4.6; coarse hazard-presence accuracy (88.0–89.2%) largely matched the 89% majority baseline, while fine-grained category accuracy remained below 40%, risk-timing accuracy was 1.6–12.4%, and exact defensive-action matching stayed below 2%.

DINOv2 Feature Distillation for Recurrent Local-Update Classifiers

2026

- Studied whether frozen DINOv2 representations can improve compact recurrent and local-update image classifiers through feature-level knowledge distillation.
- Compared the distilled recurrent model with a compact feed-forward baseline under controlled training settings.

Selective Polyp Segmentation under Dataset Variation

2026

- Studied single-pass uncertainty estimation and selective prediction for medical image segmentation under external dataset variation.

TECHNICAL SKILLS

Programming	Python, C++
ML / Vision	PyTorch, Hugging Face Transformers, PEFT/LoRA, scikit-learn; computer vision, vision-language models, multimodal learning, knowledge distillation, model evaluation
Research Engineering	Git, Docker, Kaggle; reproducible evaluation pipelines, controlled ablations, experiment tracking
Backend / Data	FastAPI, REST APIs, MySQL, PostgreSQL, SQLite

HONORS, LEADERSHIP, AND INNOVATION

- Champion, University Innovation Hub Program, IC-6 cohort; Founder and CTO, GreenLoop.
- Champion, Hult Prize at RUET, 2025; represented RUET at the Hult Prize National, 2025.
- 6th place, Inter-University Capture the Flag Competition, BUET CSE FEST, 2024.
- Top 25 finalist, BDApps Innovation Challenge, among more than 1,500 teams nationwide.

ADDITIONAL ENGINEERING

Lightning Search (Khoj – The Search)

- Built a fuzzy-search system for 10,000+ entries using n-gram indexing, candidate filtering, and similarity-based ranking with Python, FastAPI, and RapidFuzz.